

Project Roadmap for Social Navigation Planner Development and Validation on Jackal

Objective

The goal of this project is to establish a complete simulation-to-real-world pipeline for development, testing, and evaluation of a social navigation planner on the Clearpath Jackal platform. The project will create a robust simulator environment with dynamic pedestrian interactions, integrate the proposed planner, benchmark performance against baseline navigation approaches, and ultimately validate performance on the physical robot in structured and real-world scenarios.

Project Stages and Milestones

1. ROS2 and Jackal Base Environment Setup

Install and configure ROS2 (Jazzy), Gazebo, RViz, and the Clearpath Jackal simulator stack. Validate baseline robot functionality, sensor topics, TF tree, and autonomous navigation using stock navigation packages. At this stage, the simulator should represent a known-good working baseline and deployment workflow to the physical robot should be understood.

2. Hardware and Deployment Validation

Verify access to the physical Jackal platform, onboard compute resources, sensor stack, ROS compatibility, and deployment workflow. This stage ensures that simulator assumptions match physical hardware constraints early in development.

3. Custom Social Navigation Simulation Development

Develop simulation environments relevant to intended testing scenarios (e.g. hallways, indoor spaces, bottlenecks). Integrate pedestrian simulation models such as Social Force Model (SFM), ORCA, and other configurable crowd behaviors where possible. Establish reproducible scenario generation with varying crowd density, map layouts, and random seeds.

4. Planner and Navigation Stack Integration

Integrate the proposed planner into the ROS2 stack, including inference (e.g. ONNX deployment if applicable), navigation interfaces, and real-time control loop validation. Ensure compatibility with sensor topics, pedestrian information, and robot command interfaces. Verify that real-time performance requirements are met in simulation.

5. Simulation Testing and Benchmark Evaluation

Build an automated evaluation framework to collect navigation performance metrics such as:

- success rate and time-to-goal

- path efficiency and smoothness
- collision / near-collision rates
- minimum pedestrian distance
- planner latency and loop frequency

Compare against baseline planners (e.g. Nav2 local/global planners or social navigation baselines where available).

6. Real Robot Transfer and Validation

Deploy the validated planner onto the physical Jackal and perform structured real-world experiments to confirm simulation-to-real transfer. Begin with controlled scenarios and expand toward live pedestrian interaction testing where safe and feasible.

7. Extended Demonstration and Stress Testing (Stretch Goals)

Conduct extended experiments in more unstructured environments and evaluate robustness under higher complexity scenarios. Optional extensions include real-world baseline comparisons, human detection/perception integration (e.g. YOLO-based), and longer-duration autonomous operation.

Tentative Timeline

Stage	Estimated Duration
ROS2 + Jackal baseline setup	1–2 weeks
Hardware validation	1 week
Custom environments + pedestrian simulation	1–2 weeks
Planner integration + real-time validation	1–2 weeks
Metrics + simulation benchmarking	1–2 weeks
Real robot deployment + validation	2–3 weeks
Extended testing / demonstrations	2 weeks
Total Estimated Timeline	9–14 weeks

Key Risks and Challenges

Key technical risks include:

- simulator-to-real-world perception mismatch
- planner inference latency or missed real-time deadlines
- pedestrian simulator realism limitations
- sensor / TF integration inconsistencies
- human detection and tracking instability in real-world deployment

These risks will be mitigated through early hardware validation, baseline comparisons, and progressive staged testing.

Expected Outcome

At project completion, the system should provide a validated social navigation planner capable of operating in realistic dynamic environments in simulation, with demonstrated transfer to the

physical Jackal platform and quantitative performance benchmarks against baseline navigation approaches.

Useful Resources

Nav2: <https://github.com/ros-navigation/navigation2>

Installing Clearpath Simulator: <https://docs.clearpathrobotics.com/docs/ros/tutorials/simulator/install>

PedSim ROS (pedestrian simulator, needs porting to ROS2 or find an existing compatible ROS2 package for this): https://github.com/srl-freiburg/pedsim_ros

Clearpath Jackal Documentation:

Sensors: https://docs.clearpathrobotics.com/docs/ros/api/sensors_api

Offboard PC: https://docs.clearpathrobotics.com/docs/ros/installation/offboard_pc

Tutorials/Guides: https://docs.clearpathrobotics.com/docs_robots/outdoor_robots/jackal/tutorials_jackal

Run Simulator: <https://docs.clearpathrobotics.com/docs/ros/tutorials/simulator/simulate>